have an encoded word and you have a decoded word and they're in a one-to-one relationship based on a codebook. The process of ciphering is based on calculation– you have mathematical algorithms which can scramble and unscramble messages which would require the correct key.

Modern day ciphers which work on bits didn't just come out of the blue. They were based on previous research, which in turn was based on even older research.

Simple Ciphers

Substitution Cipher

This is one of the simplest ciphers possible. The process is simple and apparent from the name - you substitute things. "But what things?" is a more important question. The most common 'thing' to be substituted is a single letter. One of the most common methods substitution ciphers use is called ROT13. Here you rotate the alphabets by 13. So 'A' becomes 'N', 'B' becomes 'O' and so on until 'M' becomes 'Z'. Since the English alphabet has only 26 letters, ROT13 makes the mapping easily reversible - i.e. 'N' becomes 'A', 'O' becomes 'B' until 'Z' becomes 'M'.

Using the cipher, the word 'HELLO' becomes 'URYYB' and 'DIGIT' becomes 'QVTVG'. Notice that 'T' was substituted with 'G' because ROT13 utilizes the same algorithm for encrypting as well as decrypting the message. Thus, it provides almost no security against analysis.

One can however utilize other rotation techniques too for substitution for single letters which may include rotation techniques other than ROT13 - for example by rotating only 3 times instead of 13. That would map 'A' to 'D', 'B' to 'E' until 'W' becomes 'Z' and again 'X' becomes 'A'. Another method that can be used is random mapping of letters, something like:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Q	F	Y	M	Z	X	L	U	A	E	I	J	V	D	C	R	N	T	H	W	P	S	G	B	K	O

In such a scene, the word 'HELLO' becomes 'UZJJC' and 'DIGIT' becomes 'MALAW'. While this is better than ROT13 where same function could be used for encrypting as well as decrypting the message, it still doesn't provide enough security against cryptanalysis because simple patterns of usage of words can easily signal the ciphering.

If in case it slipped your eye, the 'random' substitution cipher we just devised above is not much of a cipher but closer to a code. The reason resides in the fact that the mapping is completely random and for ciphering or